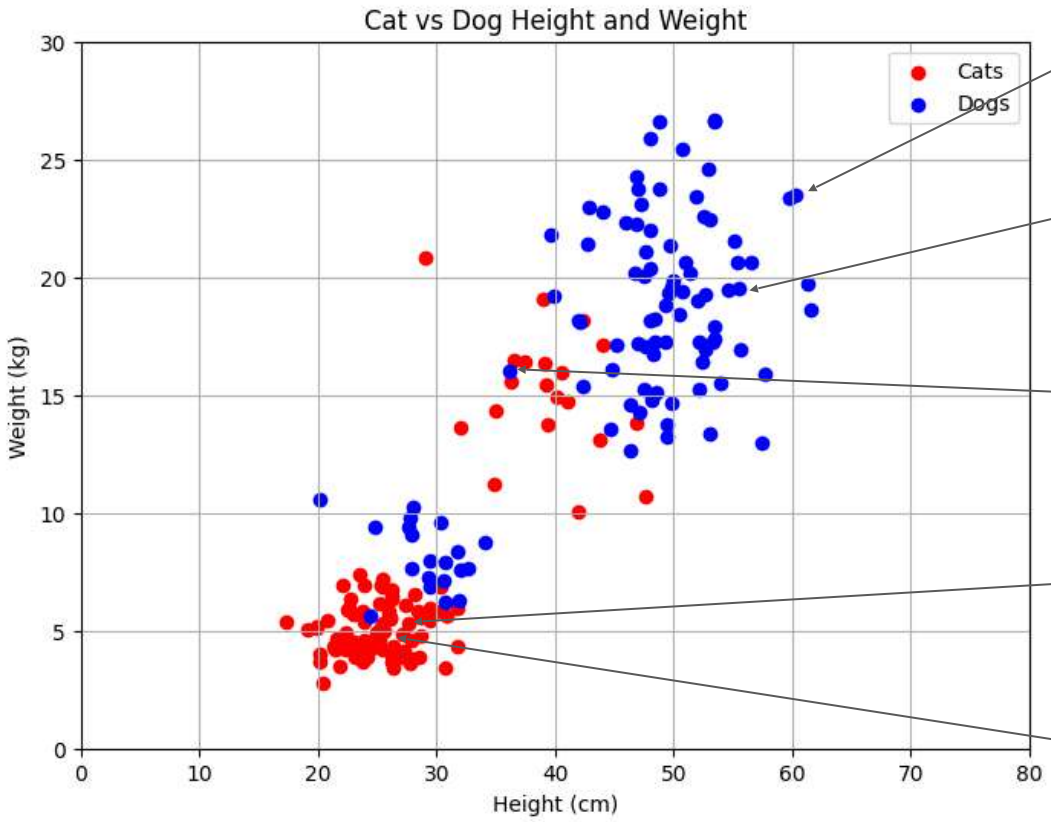
Introduction to Deep Learning

Caltech CS/EE/CMS 148a - Winter 2026

Lecture 14 - Data (II), supervision

Learning from Data

Learning from Data



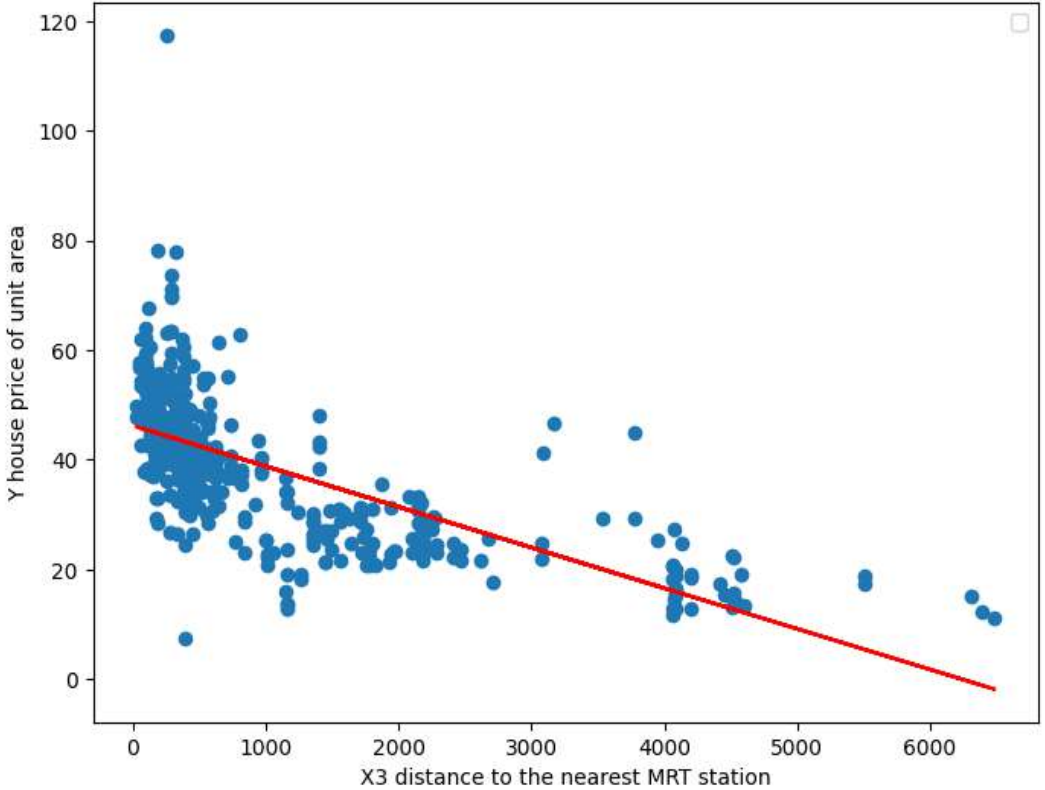
Plots are generated with Google Colab, images are from Pexels, Dots up

Regression

$$f : X \in \mathbb{R}^m \rightarrow Y \in \mathbb{R}^n$$

X1 transaction date	X2 house age	X3 distance to the nearest MRT station	X4 co
2012.917	32.0	84.87882	
2012.917	19.5	306.59470	
2013.583	13.3	561.98450	
2013.500	13.3	561.98450	
2012.833	5.0	390.56840	

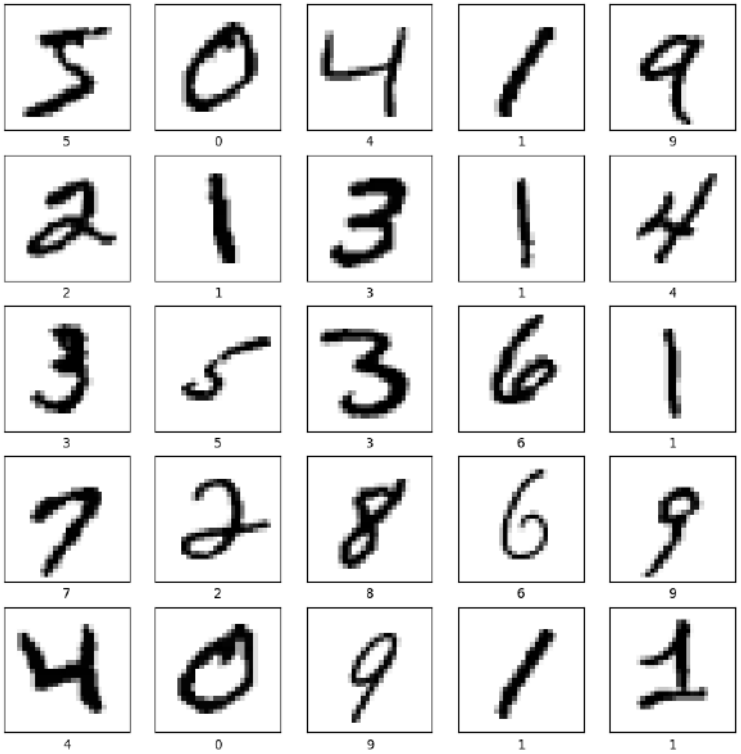
...



Real estate price prediction

Classification

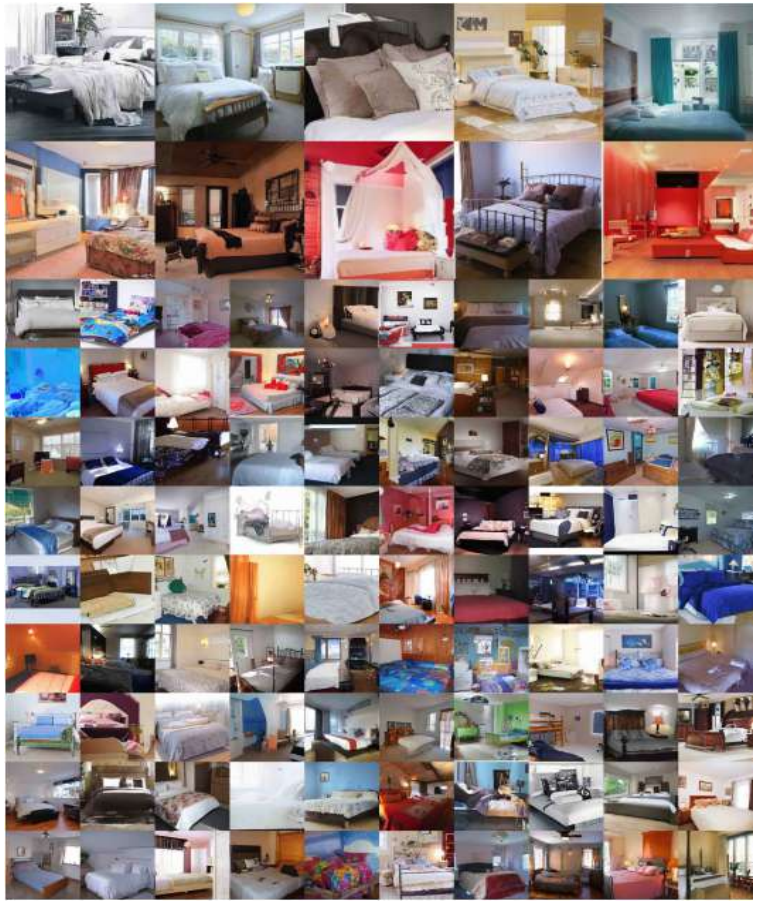
$$f : X \in \mathbb{R}^n \rightarrow Y \in \{C_1, C_2, \ldots, C_m\}$$



MNIST handwritten digit database, Yann LeCun, Corinna Cortes and Chris Burges

Probability density estimation

$$\{X_i\}_{i \in I} \in \mathbb{R}^n \to \mu_X$$



[2006.11239] Denoising Diffusion Probabilistic Models

Key elements of learning from data

- Task
- Data
- Class of functions
- Selection criterion
- Evaluation

Data

Input-output pairs that fit the task.

$$\mathcal{D} = \{x_i, y_i\}$$

X1 transaction date	X2 house age	X3 distance to the nearest MRT station	X4 number of convenience stores	X5 latitude	X6 longitude	Y house price of unit area
2012.917	32.0	84.87882	10	24.98298	121.54024	37.9
2012.917	19.5	306.59470	9	24.98034	121.53951	42.2
2013.583	13.3	561.98450	5	24.98746	121.54391	47.3
2013.500	13.3	561.98450	5	24.98746	121.54391	54.8
2012.833	5.0	390.56840	5	24.97937	121.54245	43.1

Input

Output

Class of functions

Which functions do I want to consider as candidates to solve the task?

$$\boxed{f \in \mathcal{F}}$$

$$f : X \longrightarrow Y$$

Input Output

Selection criterion

How do I select the best function to solve the task, using the data I have?

Selection criterion

$$\mathcal{D} = \{x_i, y_i\}$$

How do I select the best function to solve the task, using the data I have?

Selection criterion

How do I select the best function to solve the task, using the data I have?

$$\mathcal{D} = \{x_i, y_i\}$$

$$f = \arg\min_{f \in \mathcal{F}} \mathcal{L}(f, D)$$

Selection criterion

How do I select the best function to solve the task, using the data I have?

$$\mathcal{D} = \{x_i, y_i\}$$

$$f = \arg\min_{f \in \mathcal{F}} \mathcal{L}(f, D)$$

Evaluation

Selection criterion

How do I select the best function to solve the task, using the data I have?

$$\mathcal{D} = \{x_i, y_i\}$$

$$f = \arg\min_{f \in \mathcal{F}} \mathcal{L}(f, D)$$

Evaluation

How do I check that the solution I selected fits well new, unseen data?

Selection criterion

How do I select the best function to solve the task, using the data I have?

$$\mathcal{D} = \{x_i, y_i\}$$

$$f = \arg\min_{f \in \mathcal{F}} \mathcal{L}(f, D)$$

Evaluation

How do I check that the solution I selected fits well new, unseen data?

$$\mathcal{T} = \{x_i, y_i\}$$

Selection criterion

How do I select the best function to solve the task, using the data I have?

$$\mathcal{D} = \{x_i, y_i\}$$

$$f = \arg\min_{f \in \mathcal{F}} \mathcal{L}(f, D)$$

Evaluation

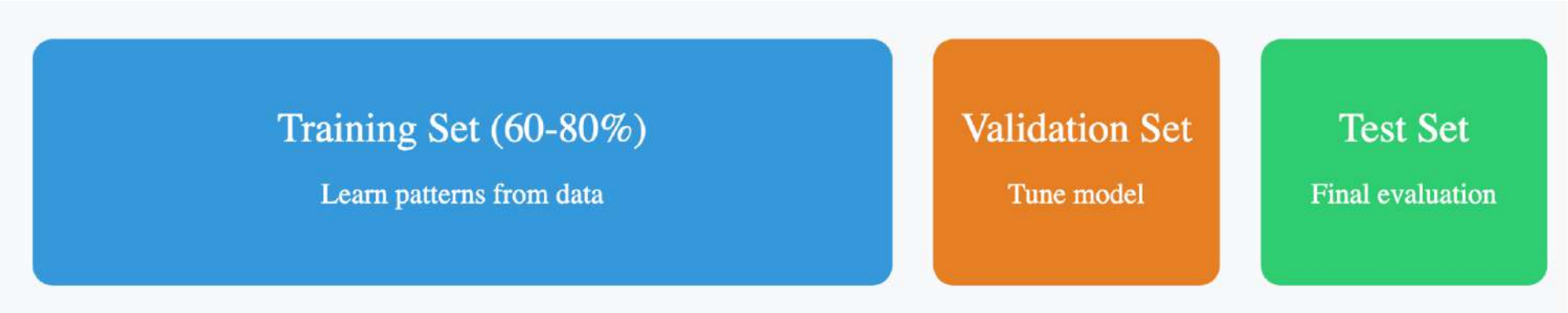
How do I check that the solution I selected fits well new, unseen data?

$$\mathcal{T} = \{x_i, y_i\}$$

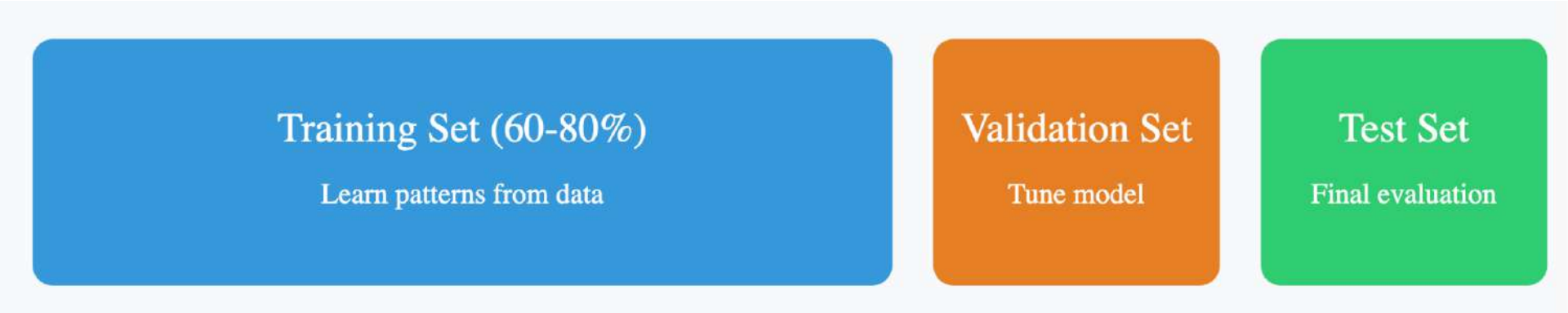
$$\mathcal{M} : (\bar{f}, \mathcal{T}) \longrightarrow \mathbb{R}$$

Training, validation, evaluation

Dataset splits

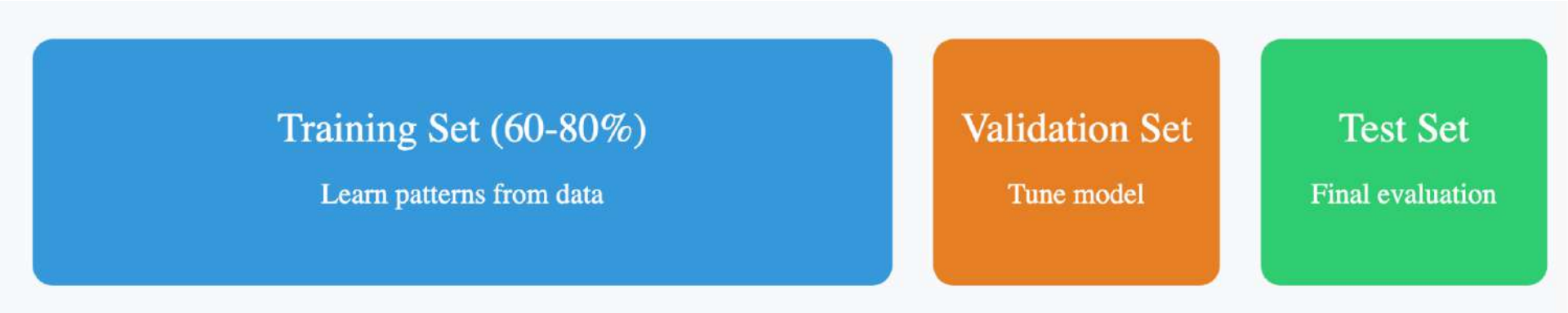


Dataset splits



- learn model weights

Dataset splits



- learn model weights
- Used repeatedly in training

Dataset splits

- learn model weights
- Used repeatedly in training

- Prevent overfitting

Dataset splits

- learn model weights
- Used repeatedly in training

- Prevent overfitting
- Select architecture

Dataset splits

- learn model weights
- Used repeatedly in training

- Prevent overfitting
- Select architecture
- Select hyperparams

Dataset splits

- learn model weights
- Used repeatedly in training

- Prevent overfitting
- Select architecture
- Select hyperparams

- Final evaluation

Dataset splits

- learn model weights
- Used repeatedly in training

- Prevent overfitting
- Select architecture
- Select hyperparams

- Final evaluation
- Report accuracy

Dataset splits

- learn model weights
- Used repeatedly in training

- Prevent overfitting
- Select architecture
- Select hyperparams

- Final evaluation
- Report accuracy
- Used only once

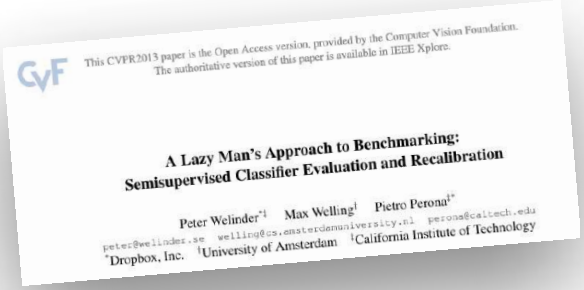
Dataset splits

- learn model weights
- Used repeatedly in training

- Prevent overfitting
- Select architecture
- Select hyperparams

- Final evaluation
- Report accuracy
- Used only once
- **Real-life test**

Out-of-sample testing



This CVPR2013 paper is the Open Access version, provided by the Computer Vision Foundation.
The authoritative version of this paper is available in IEEE Xplore.

A Lazy Man's Approach to Benchmarking: Semisupervised Classifier Evaluation and Recalibration

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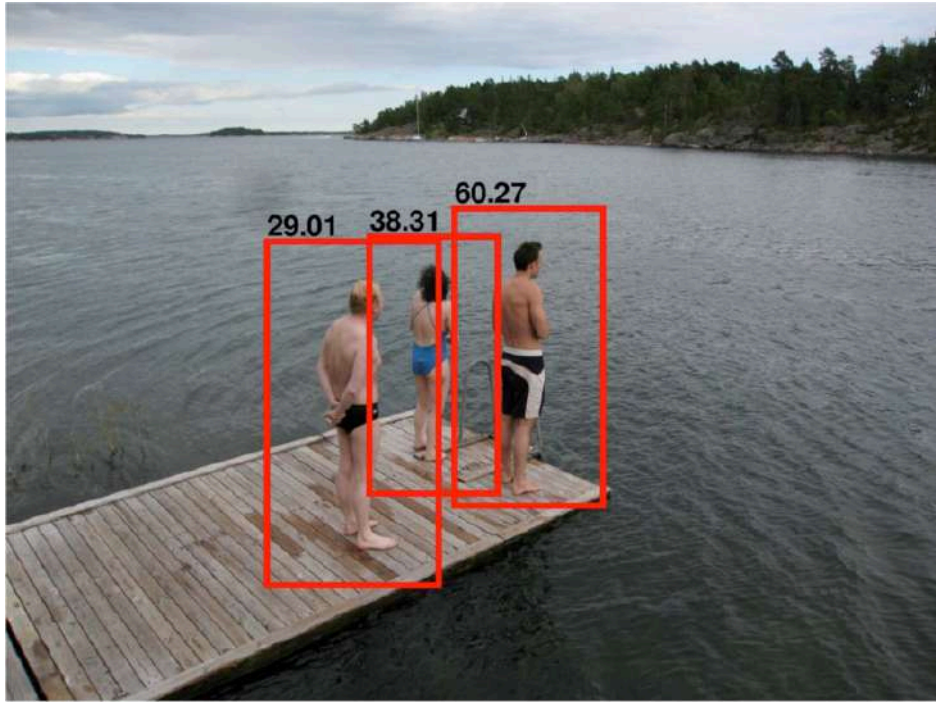
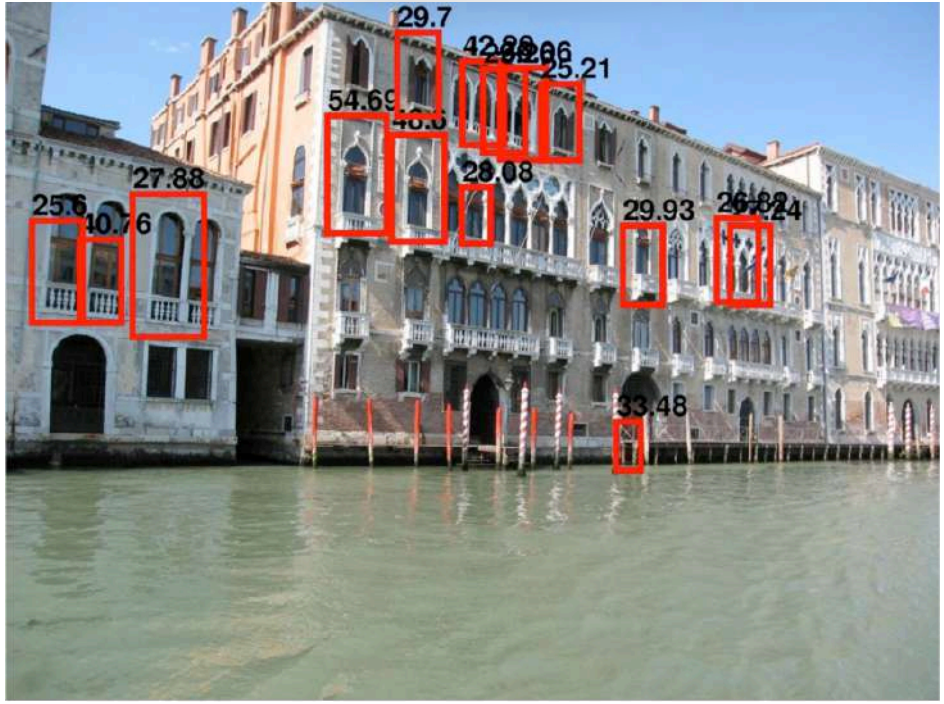
[*]Dropbox, Inc. [‡]University of Amsterdam [†]California Institute of Technology





Which data is used to determine …

	Model Hyperparameters	Reported Model Accuracy	Model weights	Model Architecture
Training				
Validation				
Evaluation				

Which data is used to determine …

	Model Hyperparameters	Reported Model Accuracy	Model weights	Model Architecture
Training				
Validation				
Evaluation				

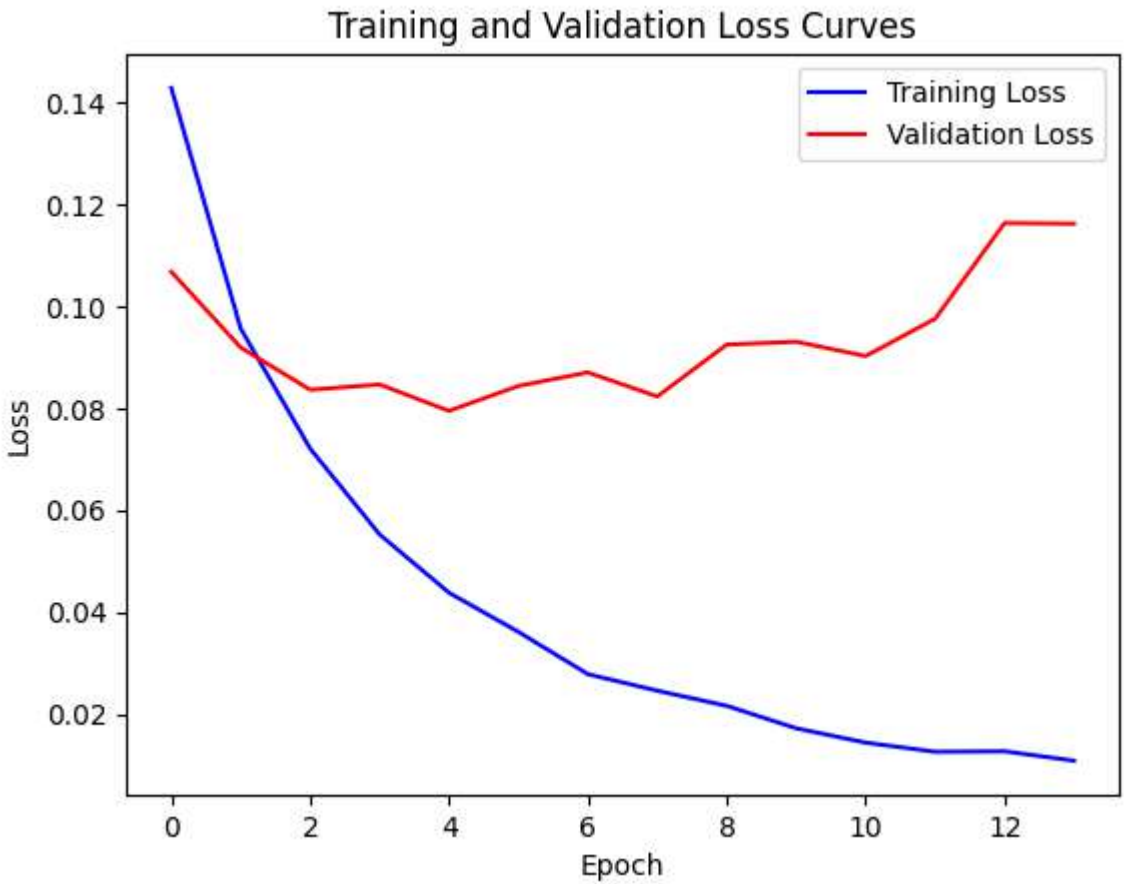
How often do we use each data segment

	Many times	Once	A small amounts of times
Training			
Validation			
Evaluation			

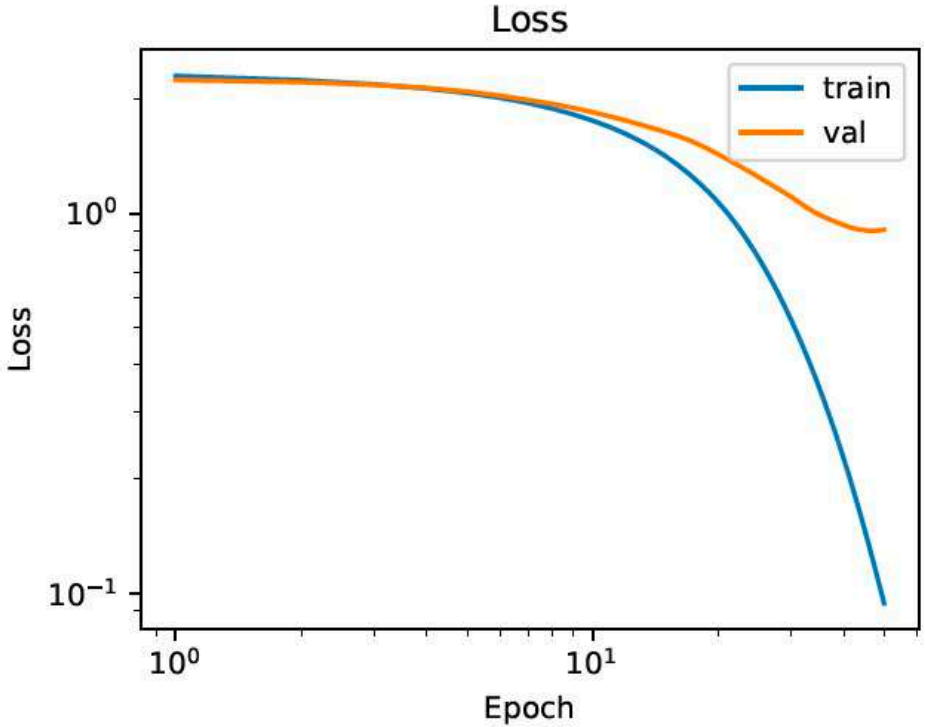
How often do we use each data segment

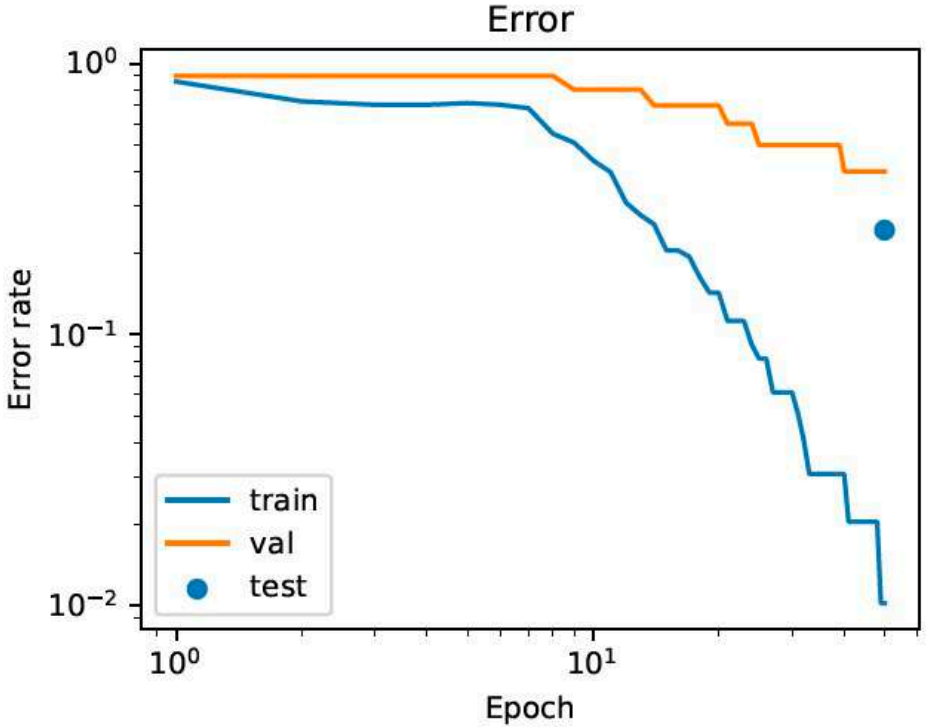
	Many times	Once	A small amounts of times
Training			
Validation			
Evaluation			

Why separate datasets?

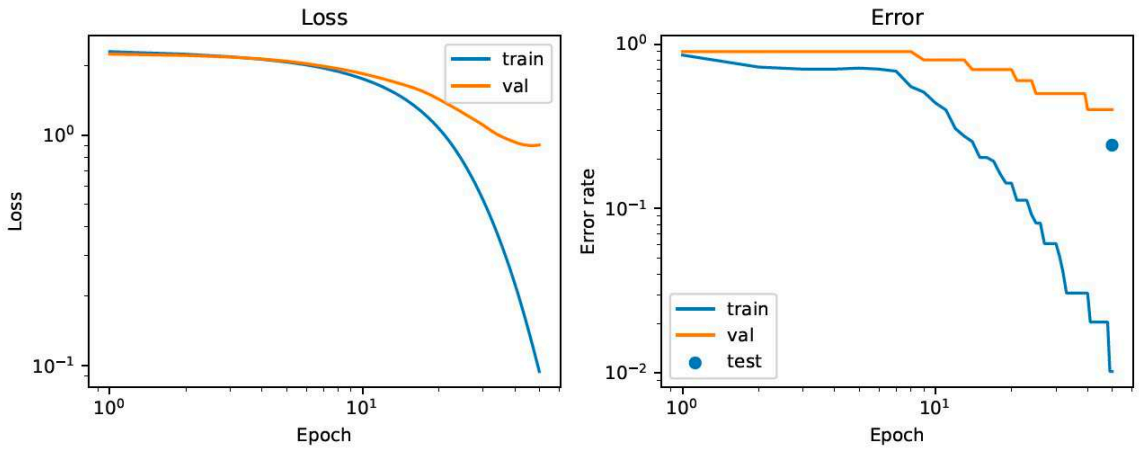


1-16-64 n=98

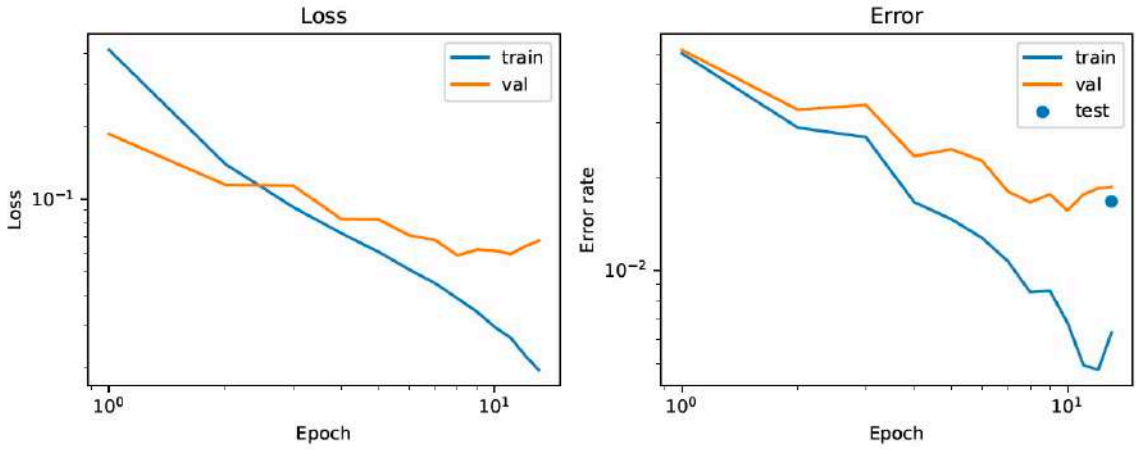

Loss
train
val
10^0
10^{-1}
Loss
10^0
10^1
Epoch

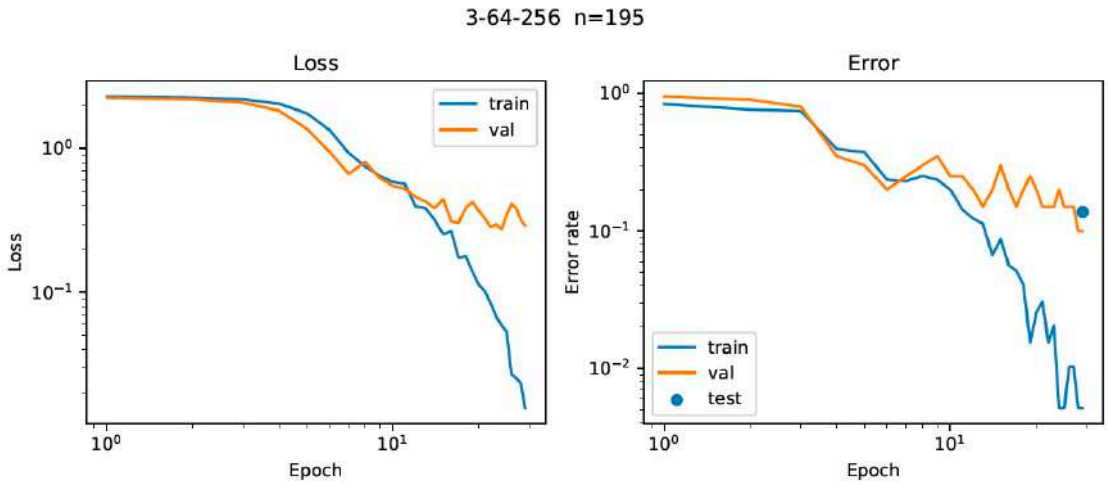

Error
10^0
10^{-1}
10^{-2}
Error rate
train
val
test
10^0
10^1
Epoch

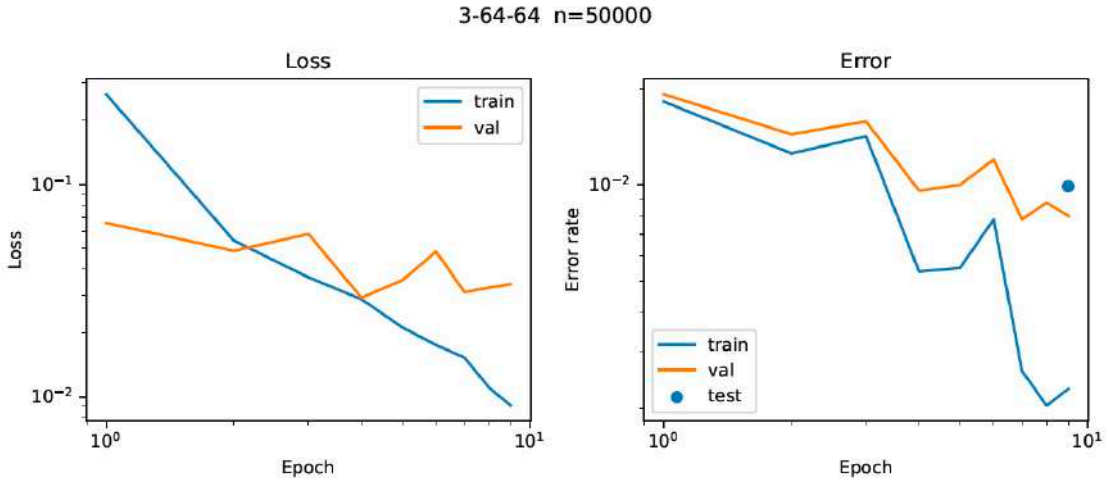
1-16-64 n=98

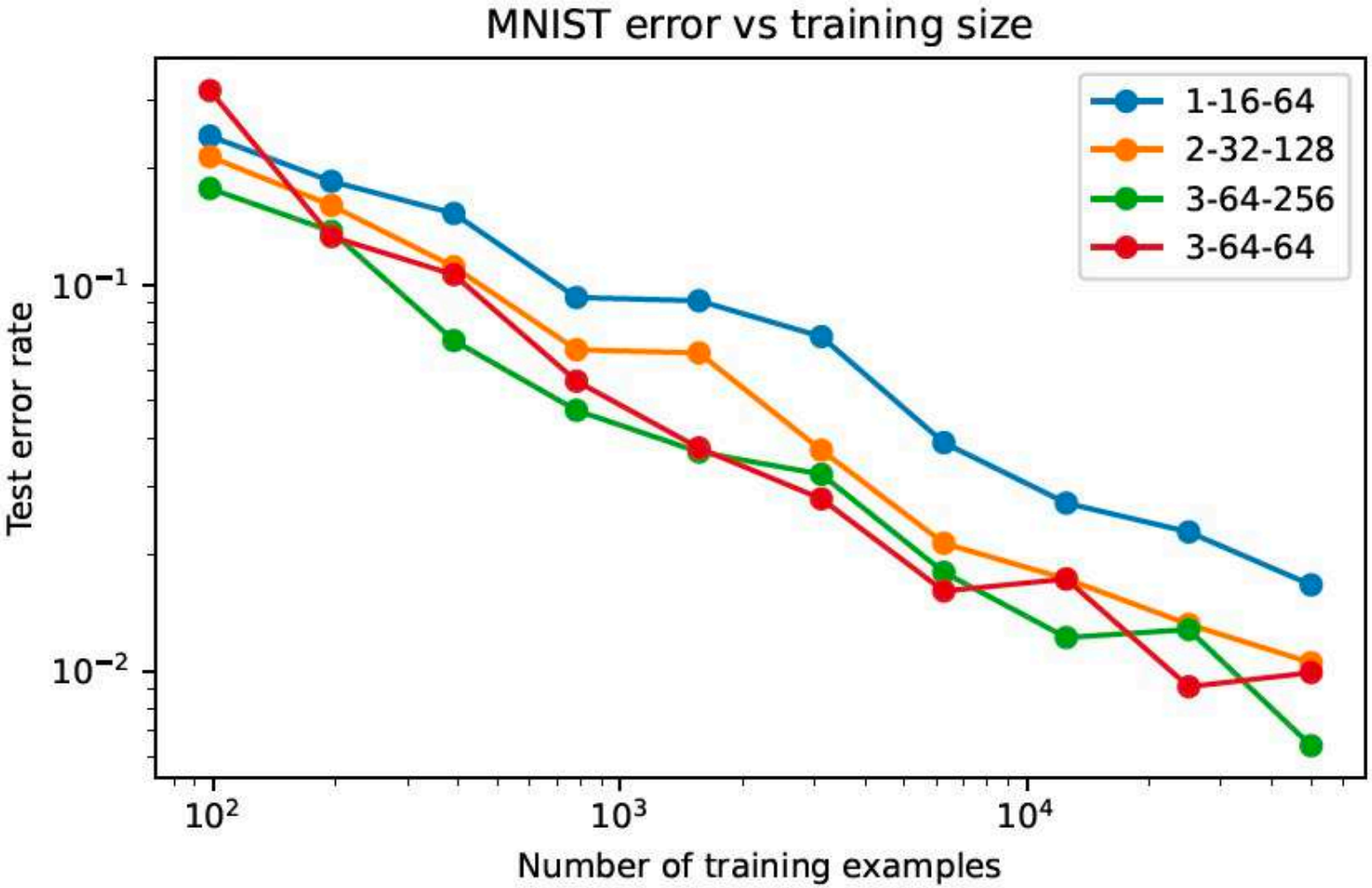


1-16-64 n=50000




3-64-256 n=195
Loss
Error
train
val
test
Loss
Error rate
Epoch
Epoch


3-64-64 n=50000
Loss
Error
train
val
test
Loss
Error rate
Epoch
Epoch

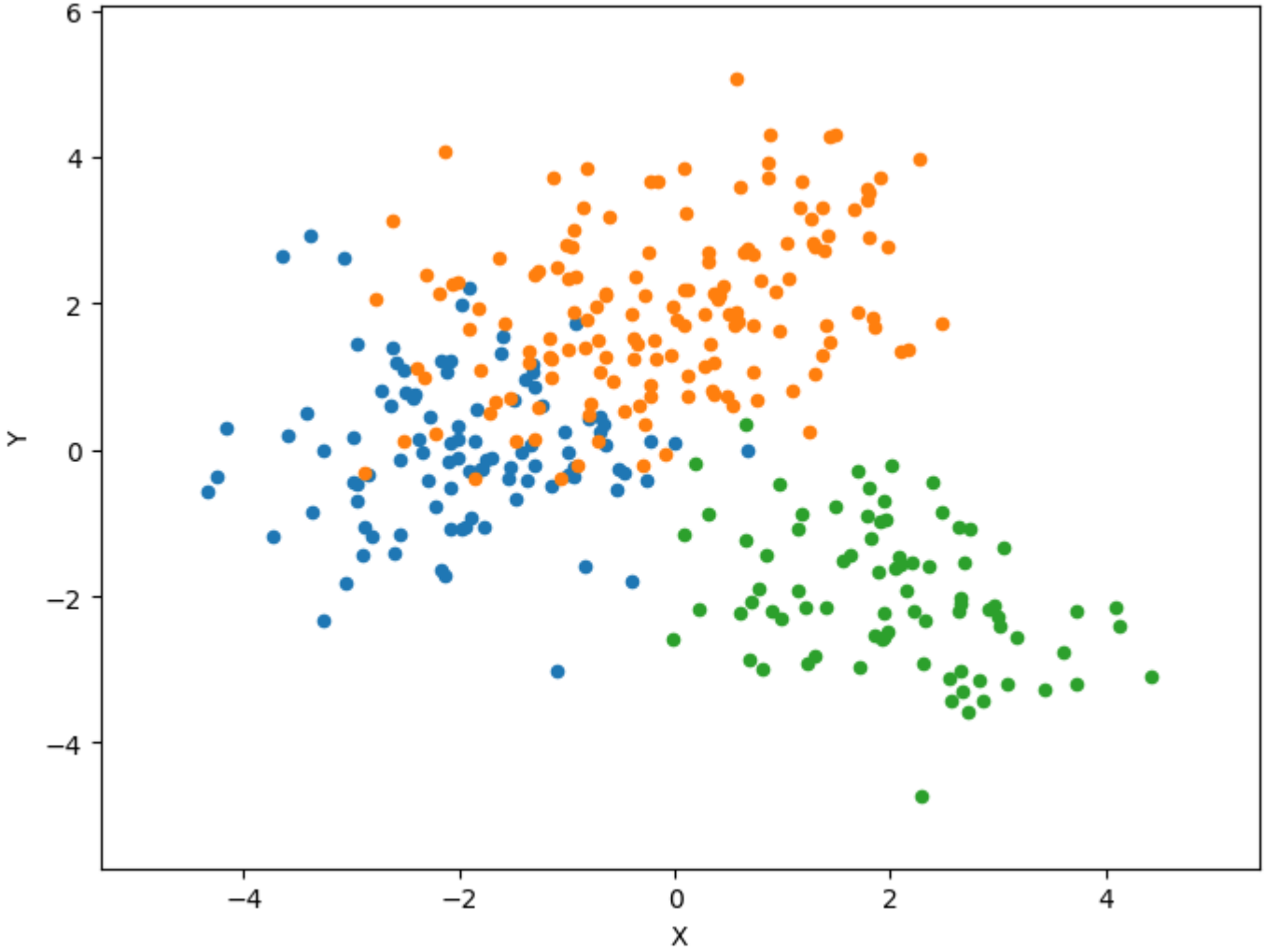

MNIST error vs training size
1-16-64
2-32-128
3-64-256
3-64-64
Test error rate
10^{-1}
10^{-2}
10^{2}
10^{3}
10^{4}
Number of training examples

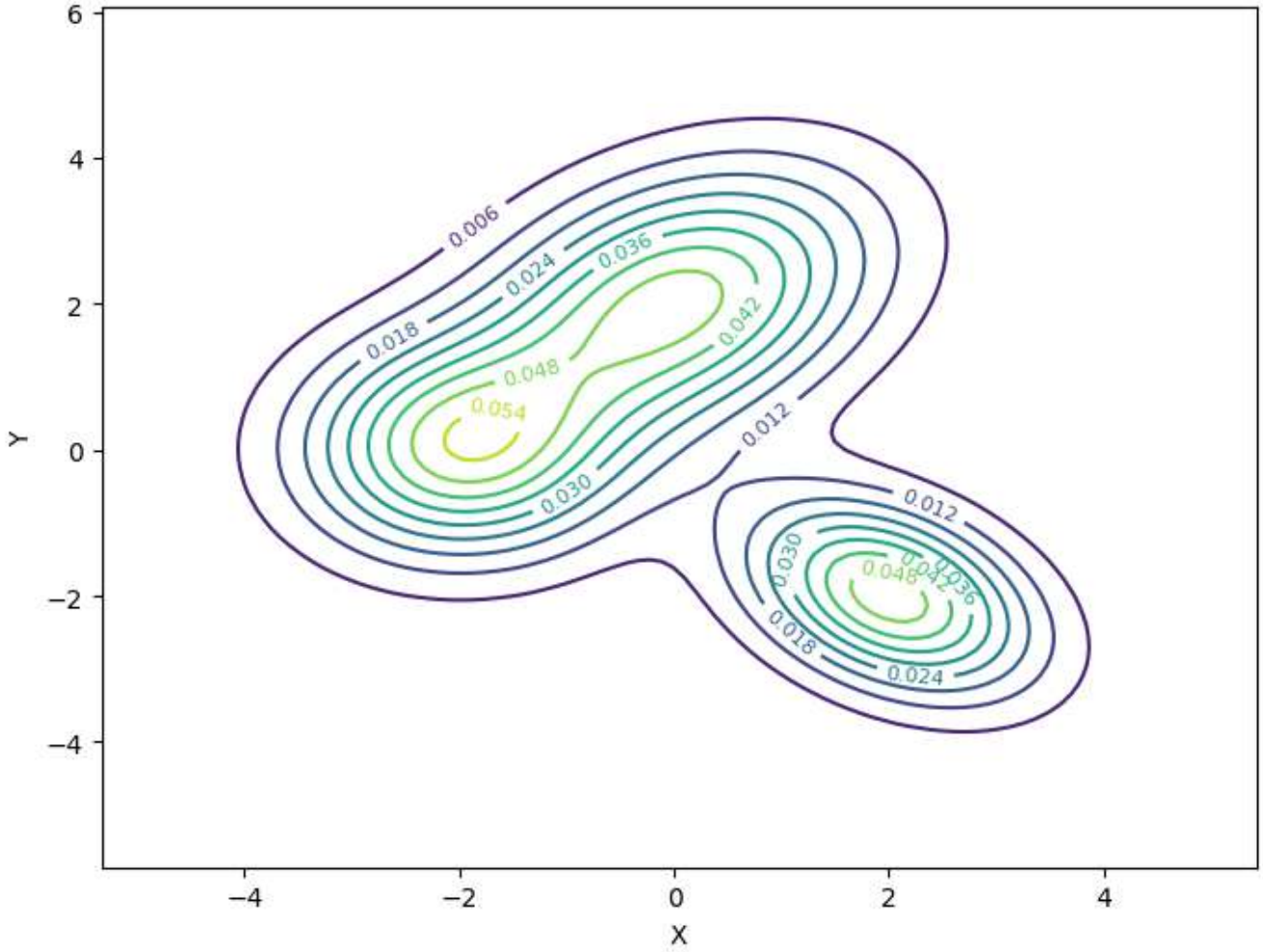
Learning and supervision

Unsupervised learning

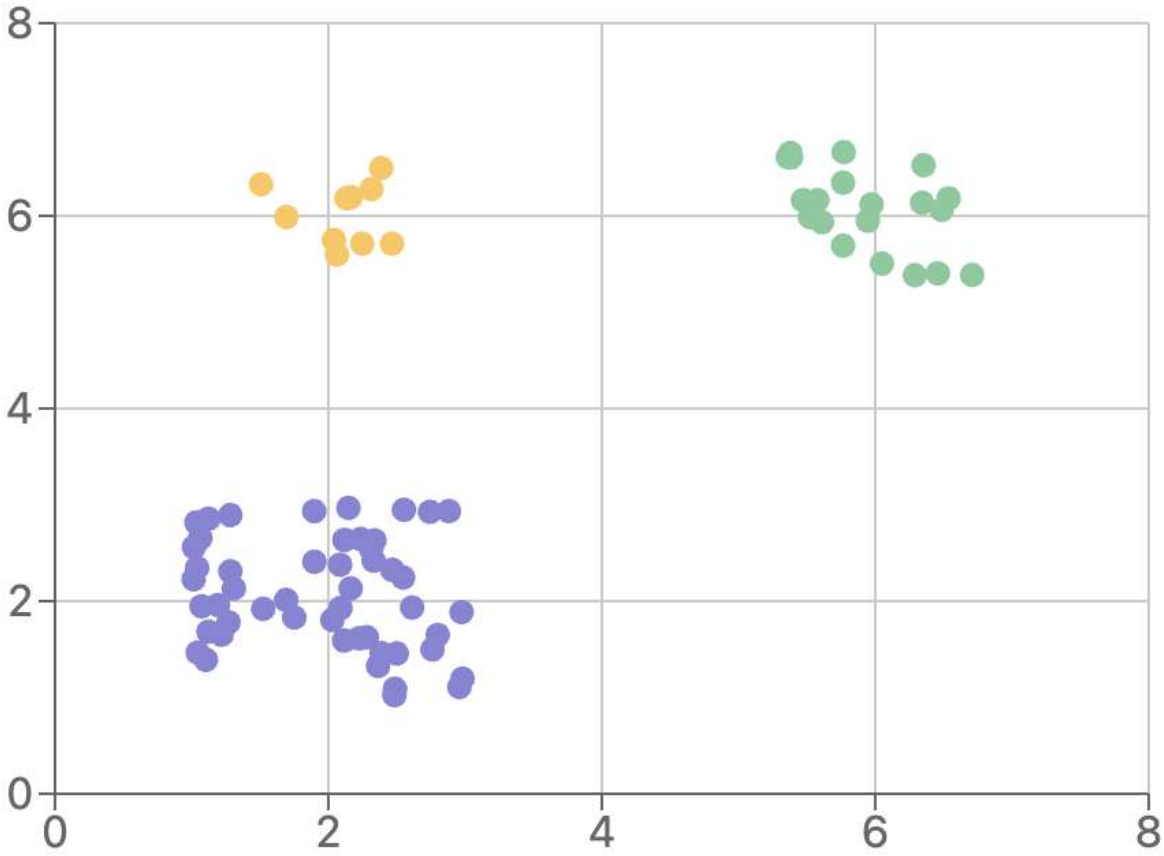
$$\{X_i\}_{i \in I} \in \mathbb{R}^n \rightarrow \mu_X$$

The task can be solved by estimating probability density of a data set

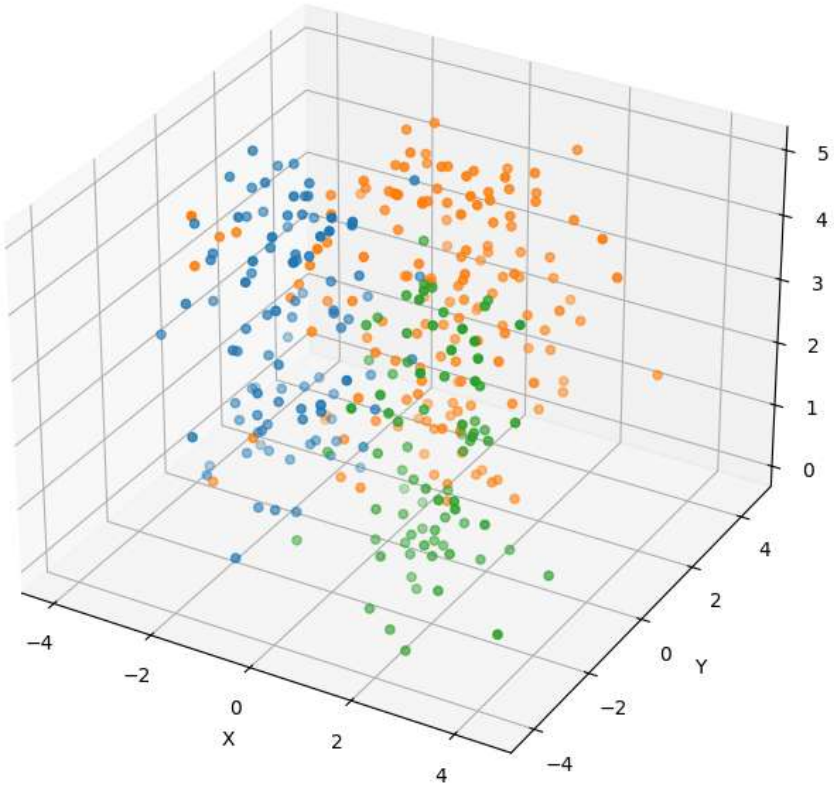

6
4
2
0
−2
−4
Y
−4
−2
0
2
4
X

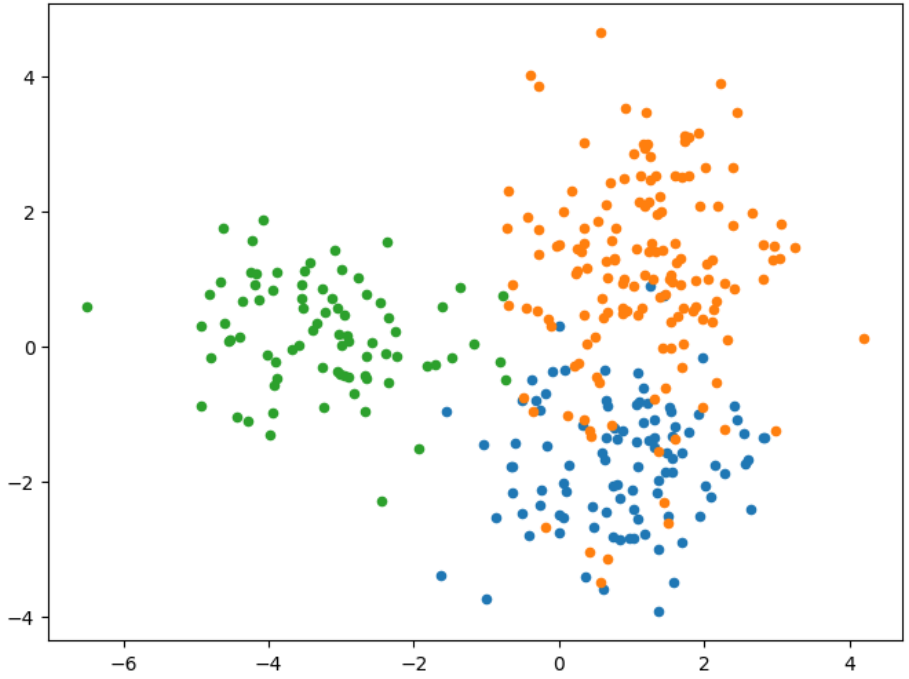

6
4
2
0
−2
−4
Y
−4
−2
0
2
4
X
0.006
0.018
0.024
0.036
0.042
0.048
0.054
0.030
0.012
0.012
0.030
0.048
0.042
0.036
0.018
0.024

Clustering

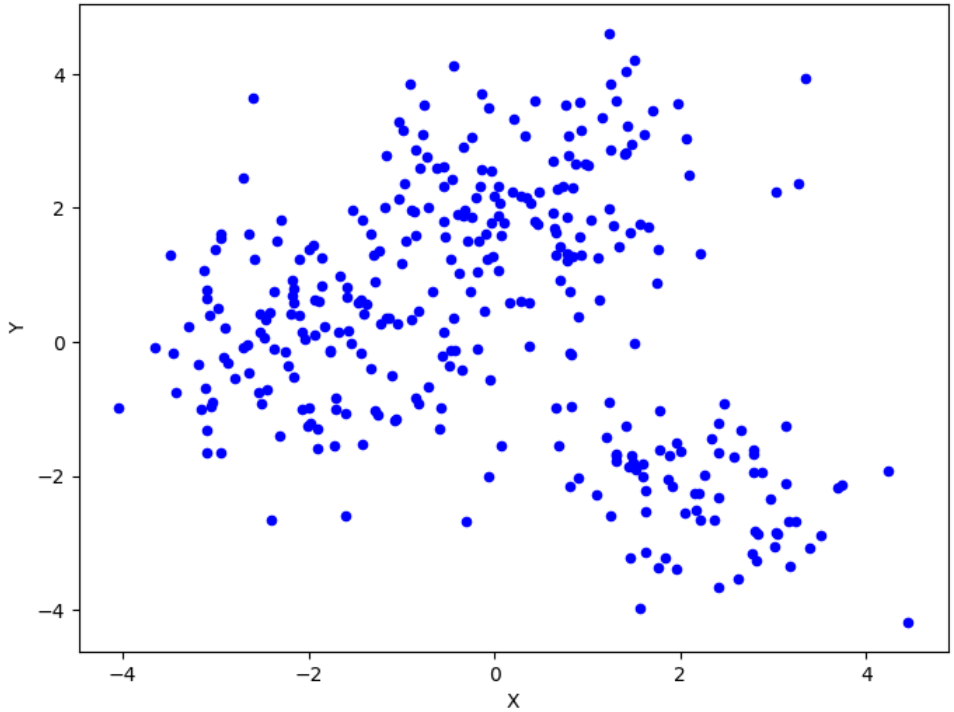


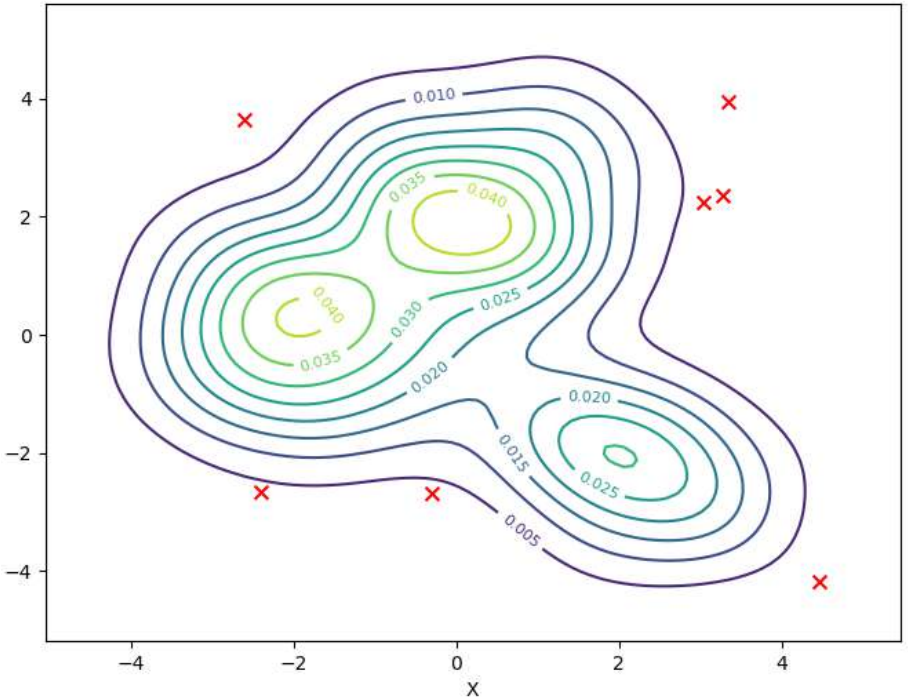
Dimensionality reduction



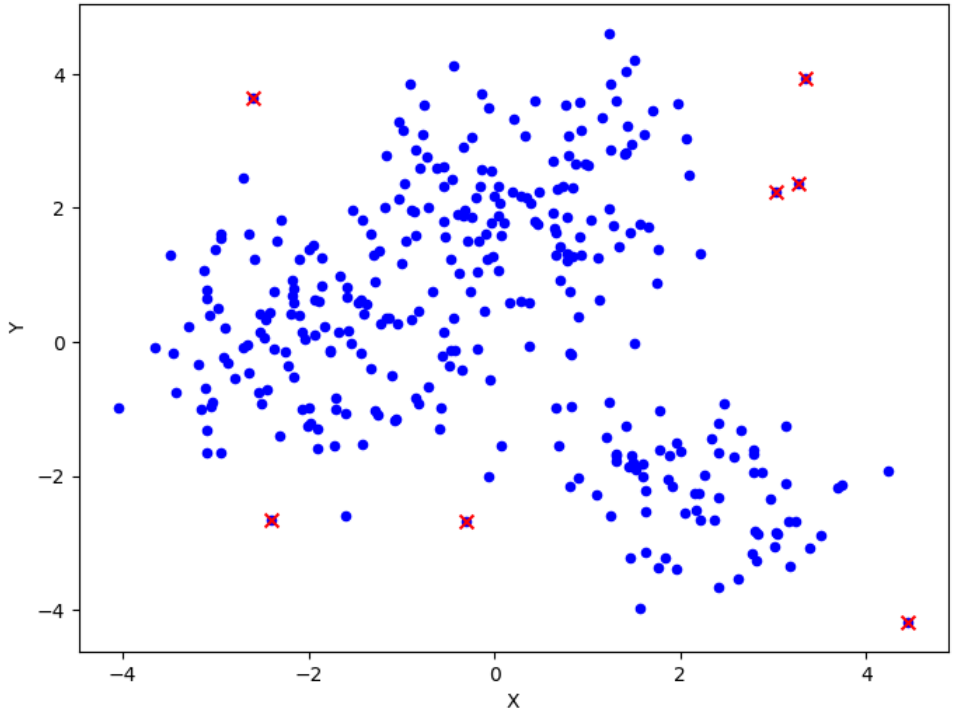


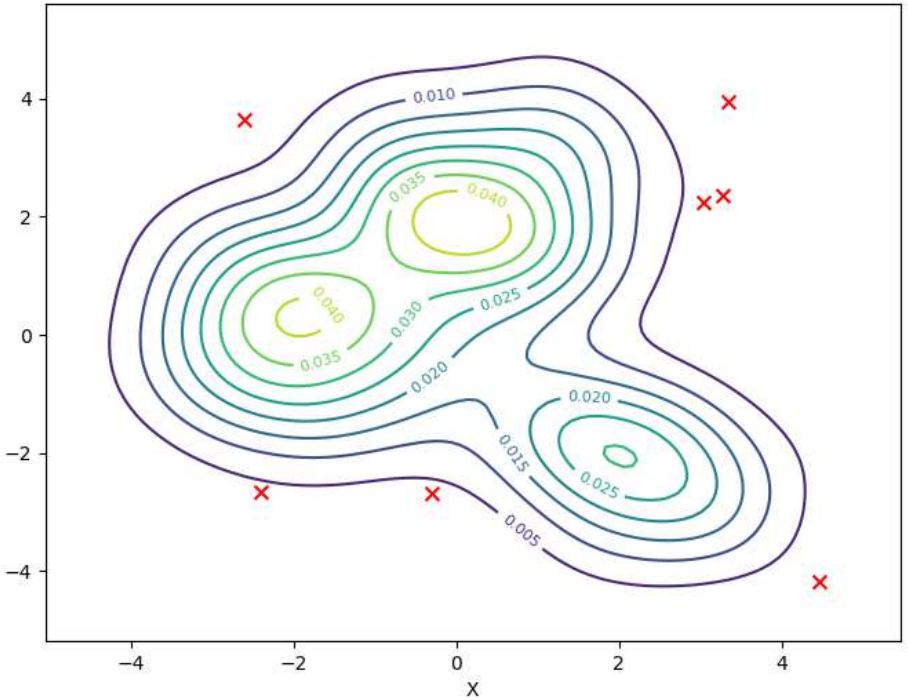
Anomaly detection





Anomaly detection





Generative models

[2006.11239] Denoising Diffusion Probabilistic Models

Supervised learning

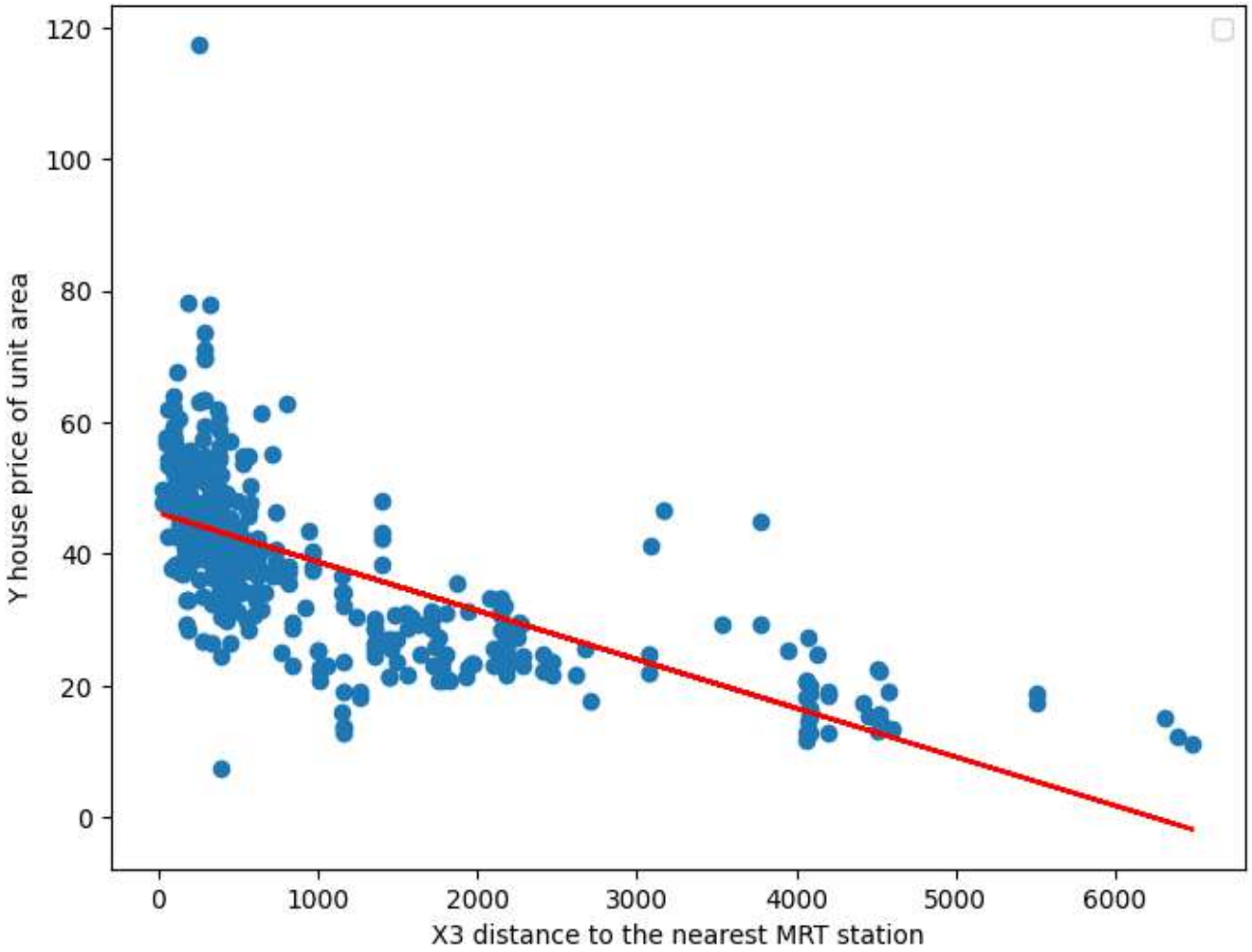
$$\mathcal{D} = \{x_i, y_i\}$$

$$f : X \in \mathbb{R}^m \rightarrow Y \in \mathbb{R}^n$$

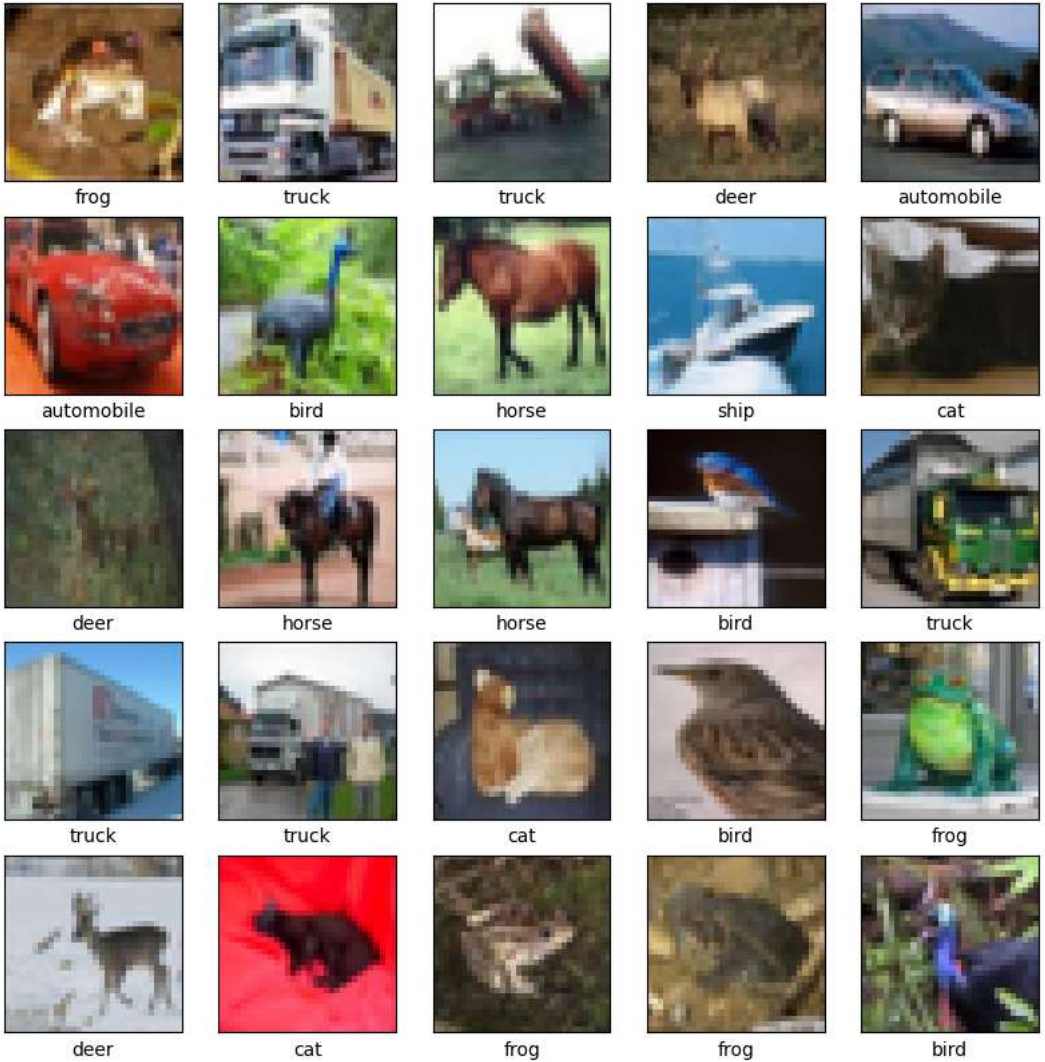
$$f : X \in \mathbb{R}^n \rightarrow Y \in \{C_1, C_2, \ldots, C_m\}$$

The data contains clear inputs and outputs for the task, and for each data point we have both available at training and validation time.

Regression


120
100
80
60
40
20
0
Y house price of unit area
0
1000
2000
3000
4000
5000
6000
X3 distance to the nearest MRT station

Classification



https://www.kaggle.com/c/cifar-10/

Sequence to sequence

"We must ensure that this does not happen again."
"Wir müssen sicherstellen, dass dies nicht wieder geschieht."

"The government has announced a new plan to tackle climate change."
"El gobierno ha anunciado un nuevo plan para abordar el cambio climático."

"The economy is expected to grow by 2% this year."
"预计今年经济将增长 2%。"

"I like to read books in the evening."
"私は夕方に本を読むのが好きです。"

"The cat sat on the mat."
"Il gatto si è seduto sul tappeto."

Semi-supervised learning

$$\mathcal{D}_{small} = \{x_i, y_i\}$$

$$\mathcal{D}_{large} = \{x_i\}$$

Semi-supervised learning

$$\mathcal{D}_{small} = \{x_i, y_i\}$$

$$\mathcal{D}_{large} = \{x_i\}$$

Labeling data could be

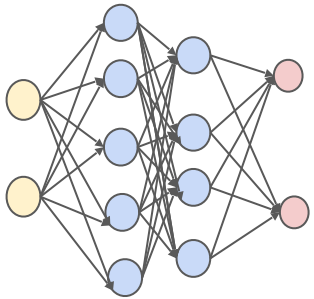
- Difficult
- Time consuming
- Expensive
- Subject to constraints
  - Privacy
  - Safety

Pseudo-labels

Self-training

1. Train a model with labeled data

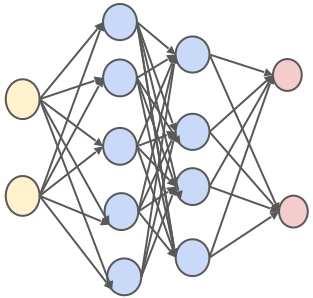
$$\mathcal{D}_{small} = \{x_i, y_i\}$$



Self-training

1. Train a model with labeled data
2. Run the model on the unlabeled data and obtain pseudo-labels

$$\mathcal{D}_{large} = \{x_i\}$$



$$\bar{\mathcal{D}}_{large} = \{x_i, \bar{y}_i\}$$

Self-training

1. Train a model with labeled data
2. Run the model on the unlabeled data and obtain pseudo-labels
3. Select a subset of pseudo-labels

$$\bar{\mathcal{D}}_{large} = \{x_i, \bar{y}_i\}$$

$$\mathcal{D}'_{small} \subset \bar{\mathcal{D}}_{large}$$

Self-training

1. Train a model with labeled data
2. Run the model on the unlabeled data and obtain pseudo-labels
3. Select a subset of pseudo-labels
    a. Confidence
    b. Diversity

$$\bar{\mathcal{D}}_{large} = \{x_i, \bar{y}_i\}$$

$$\mathcal{D}'_{small} \subset \bar{\mathcal{D}}_{large}$$

Self-training

1. Train a model with labeled data
2. Run the model on the unlabeled data and obtain pseudo-labels
3. Select a subset of pseudo-labels
    a. Confidence
    b. Diversity
4. Add the corresponding data points + pseudo labels to the labeled data

$$\mathcal{D}_{small} = \mathcal{D}_{small} \cup \mathcal{D}'_{small}$$

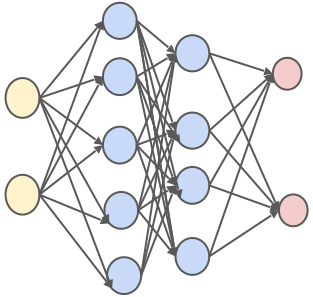
$$\mathcal{D}_{large} = \mathcal{D}_{large} \setminus \mathcal{D}'_{small}$$

Self-training

1. Train a model with labeled data
2. Run the model on the unlabeled data and obtain pseudo-labels
3. Select a subset of pseudo-labels
    a. Confidence
    b. Diversity
4. Add the corresponding data points + pseudo labels to the labeled data
5. Repeat from 2 until validation meets criteria

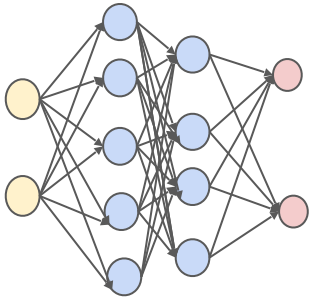
$$\mathcal{D}_{small} = \{x_i, y_i\}$$

Train



$$\mathcal{D}_{large} = \{x_i\}$$

Evaluate



$$\bar{\mathcal{D}}_{large} = \{x_i, \bar{y}_i\}$$

$$\mathcal{D}'_{small} \subset \bar{\mathcal{D}}_{large}$$

$$\mathcal{D}_{small} = \mathcal{D}_{small} \cup \mathcal{D}'_{small}$$

$$\mathcal{D}_{large} = \mathcal{D}_{large} \setminus \mathcal{D}'_{small}$$

Pseudo-label (Lee et al., 2014)

$$L = \frac{1}{n}\sum_{m=1}^{n}\sum_{i=1}^{C} L(y_i^m, f_i^m) + \alpha(t)\frac{1}{n'}\sum_{m=1}^{n'}\sum_{i=1}^{C} L(y_i'^m, f_i'^m),$$

Pseudo-label: The **simple** and **efficient semi-supervised learning** method for **deep** neural networks

Assumptions

- The initial model is correct
- Similar features yield similar predictions
- Decision boundary lie in regions with low density
- The function is smooth

The New York Times

TheUpshot

When A.I.'s Output Is a Threat to A.I. Itself

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By Aatish Bhatia

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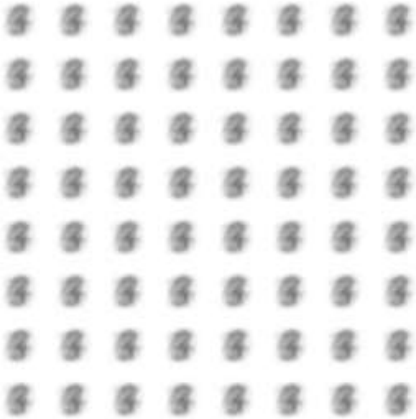
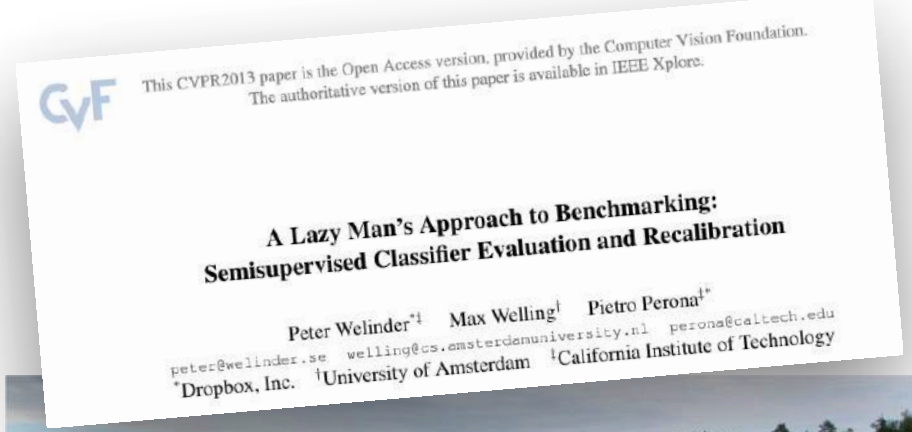
MNIST

Synthetic



Final

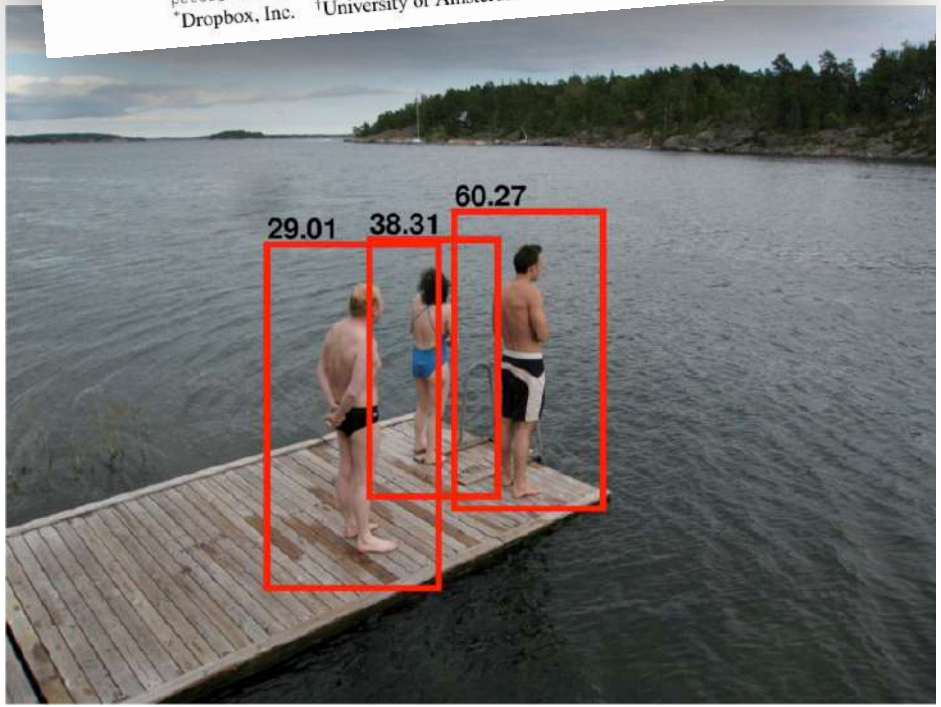
Semi-supervised evaluation

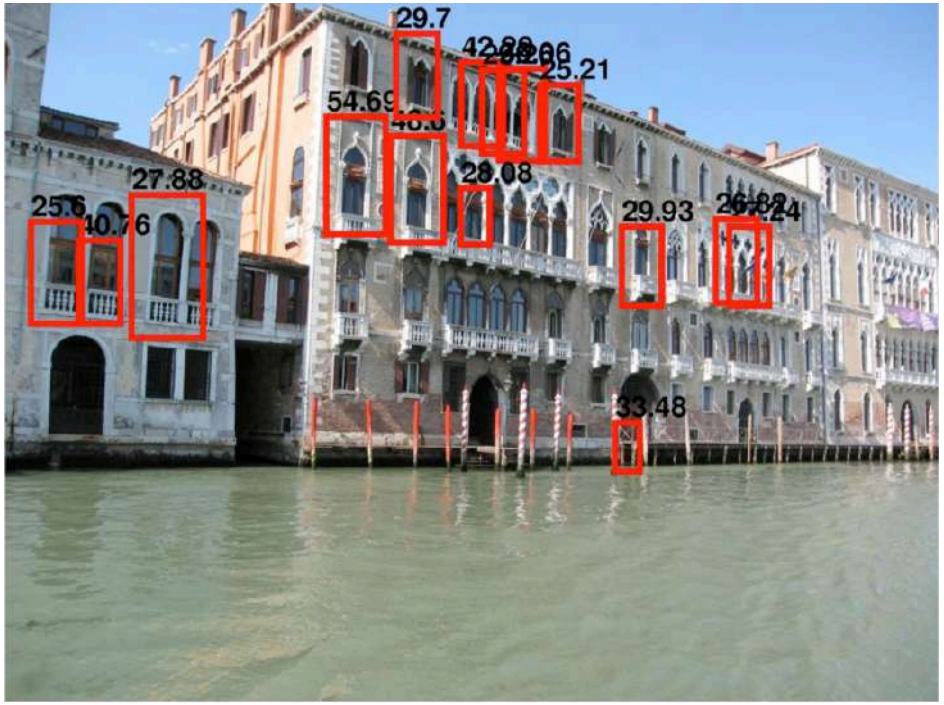


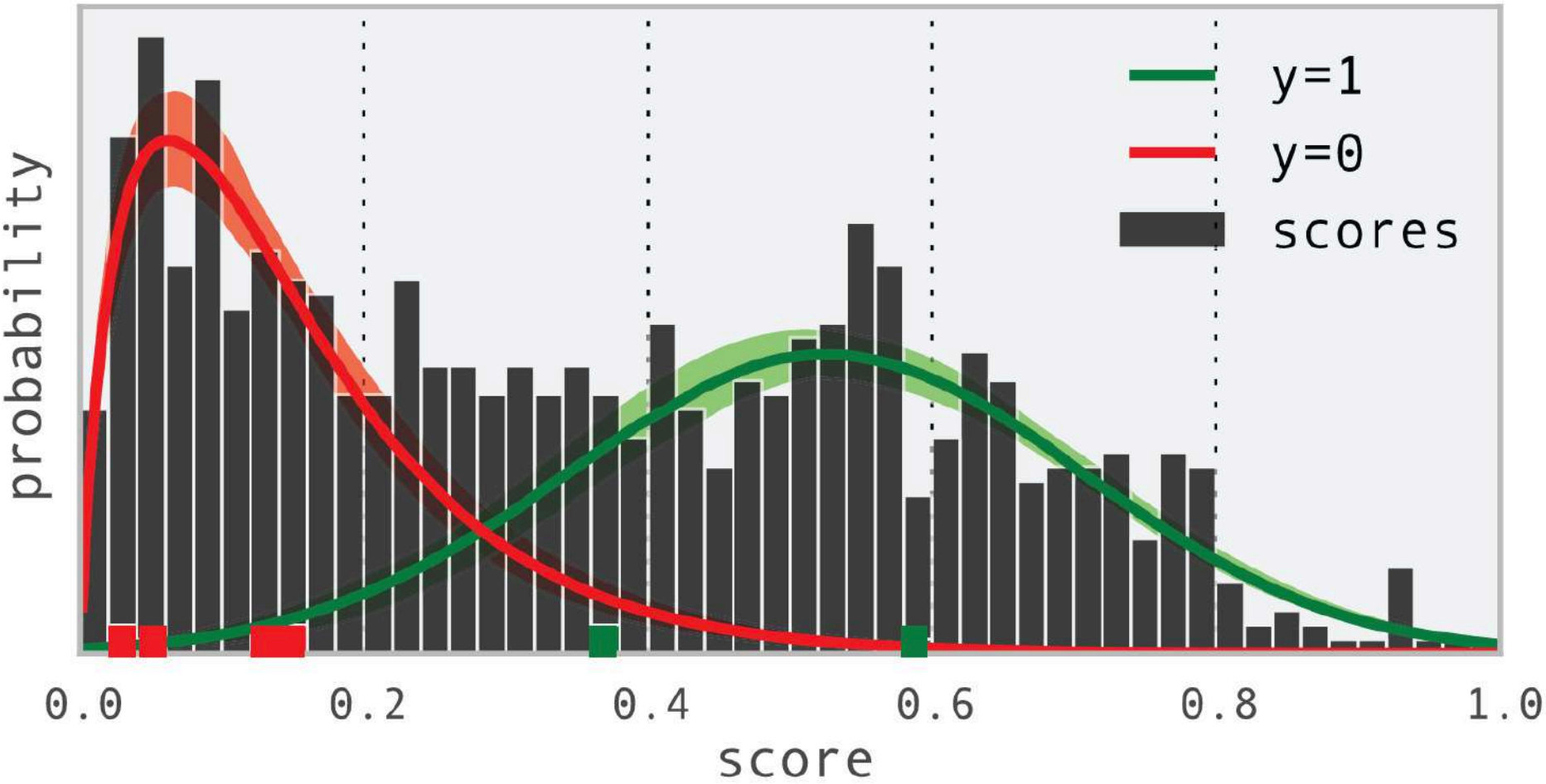
This CVPR2013 paper is the Open Access version, provided by the Computer Vision Foundation.
The authoritative version of this paper is available in IEEE Xplore.

A Lazy Man's Approach to Benchmarking: Semisupervised Classifier Evaluation and Recalibration

Peter Welinder*‡ Max Welling† Pietro Perona‡*
peter@welinder.se welling@cs.amsterdamuniversity.nl perona@caltech.edu
*Dropbox, Inc. †University of Amsterdam ‡California Institute of Technology



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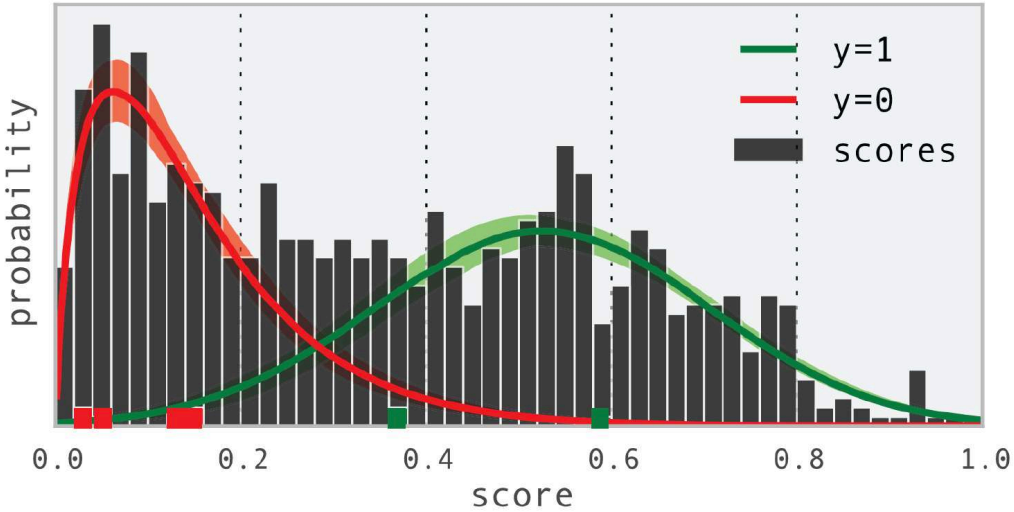
0.2
0.4
0.6
0.8
1.0
score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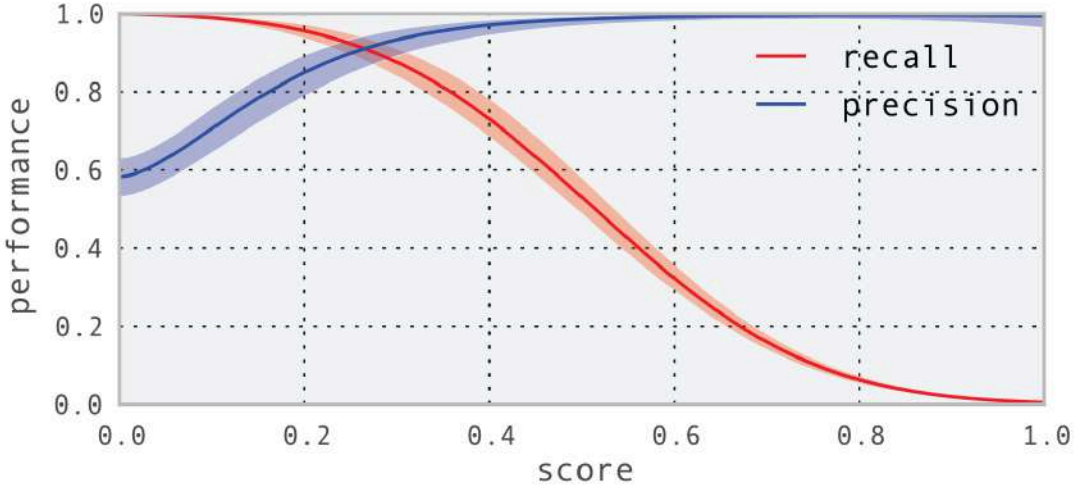
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precision
performance
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0.8
0.6
0.4
0.2
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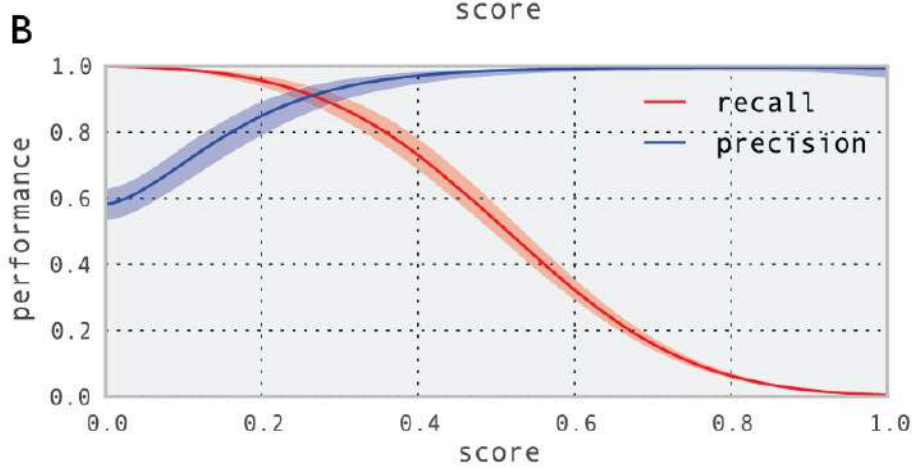
0.0
0.2
0.4
0.6
0.8
1.0
score

A
y=1
y=0
scores
probability
0.0
0.2
0.4
0.6
0.8
1.0
score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recall
precision
performance
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0.6
0.4
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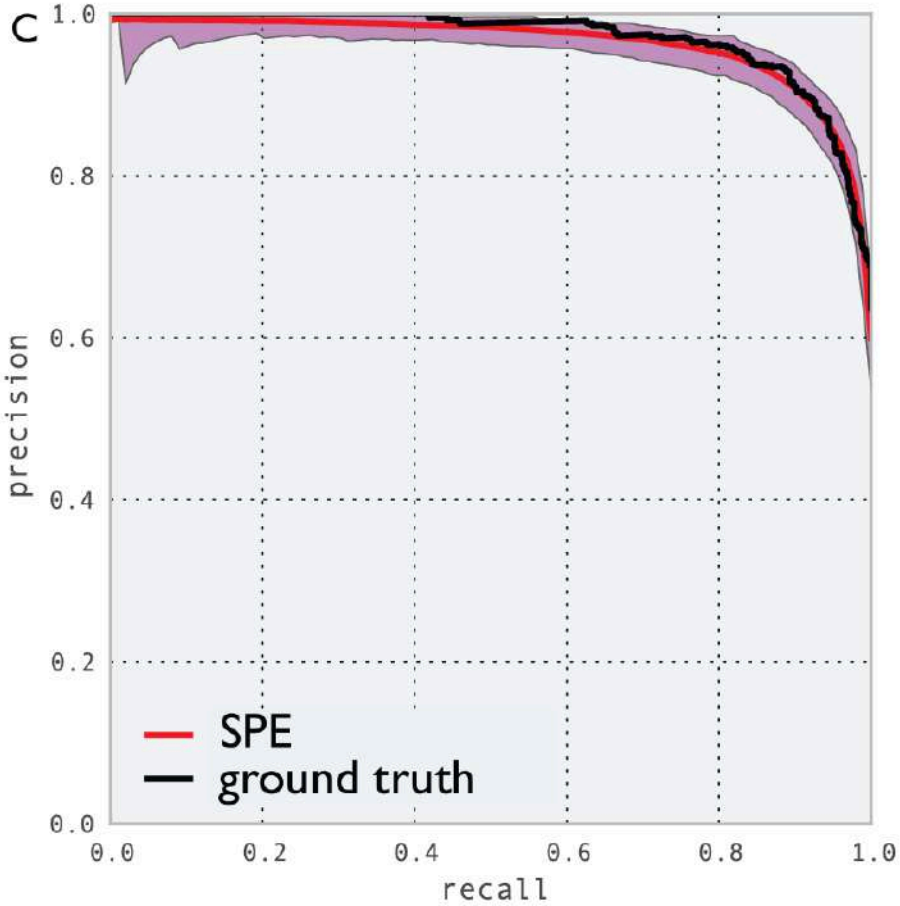
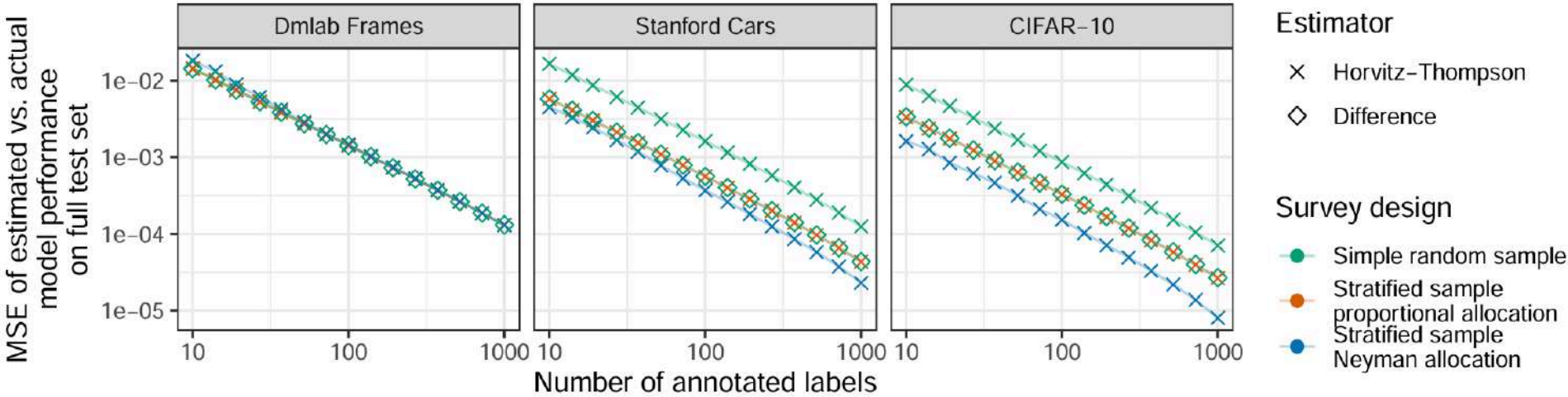
0.0
score


C
SPE
ground truth
precision
recall
1.0
0.8
0.6
0.4
0.2
0.0

A Framework for Efficient Model Evaluation through Stratification, Sampling, and Estimation

Riccardo Fogliato[†] Pratik Patil[‡] Mathew Monfort[†] Pietro Perona[†§]



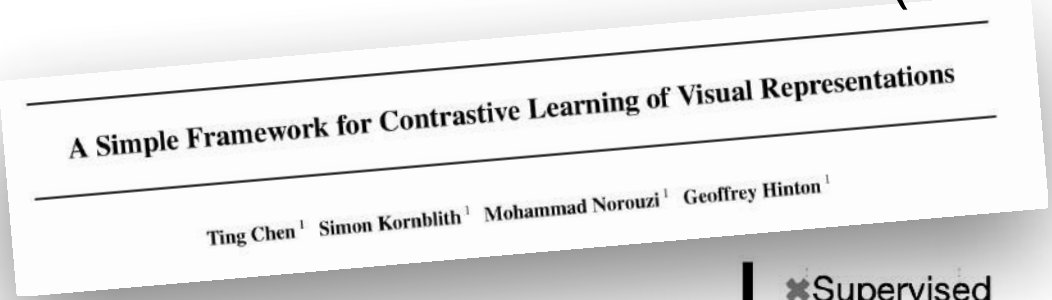
Consistency regularization

Consistency regularization

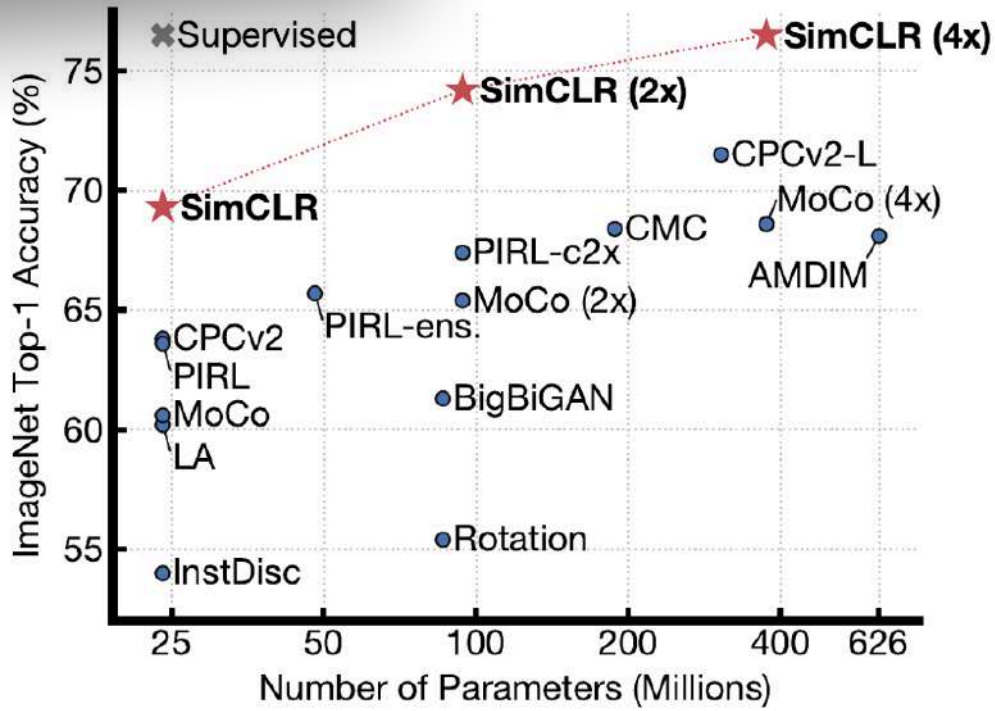
Enforce consistent prediction for similar inputs

Add a consistency component to the loss function

SIMCLR - extreme case with (almost) no labels


A Simple Framework for Contrastive Learning of Visual Representations

Ting Chen [1] Simon Kornblith [1] Mohammad Norouzi [1] Geoffrey Hinton [1]



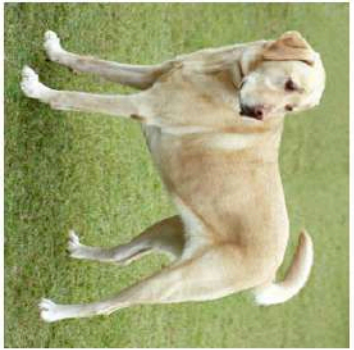
(a) Original

(b) Crop and resize

(c) Crop, resize (and flip)

(d) Color distort. (drop)

(e) Color distort. (jitter)



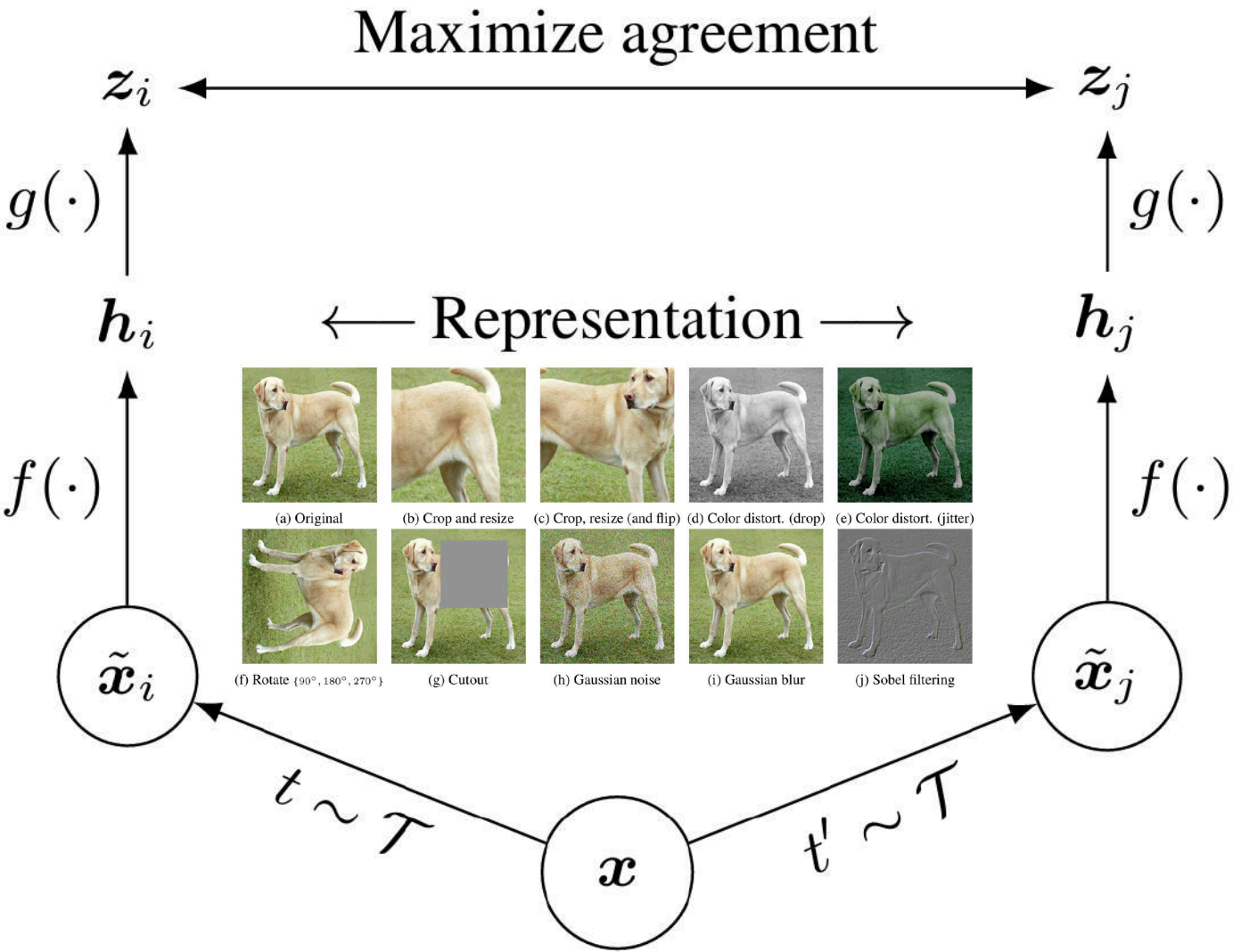
(f) Rotate $\{90°, 180°, 270°\}$

(g) Cutout

(h) Gaussian noise

(i) Gaussian blur

(j) Sobel filtering


Maximize agreement
z_i
z_j
$g(\cdot)$
$g(\cdot)$
h_i
Representation
h_j
$f(\cdot)$
$f(\cdot)$
(a) Original
(b) Crop and resize
(c) Crop, resize (and flip)
(d) Color distort. (drop)
(e) Color distort. (jitter)
(f) Rotate {90°, 180°, 270°}
(g) Cutout
(h) Gaussian noise
(i) Gaussian blur
(j) Sobel filtering
$\tilde{x}_i$
$\tilde{x}_j$
$t \sim \mathcal{T}$
$t' \sim \mathcal{T}$
x